KS3 Electricity Unit Knowledge Organiser

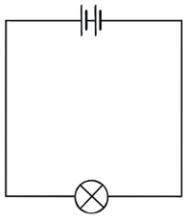
Keywords and their definitions:

voltage	The strength of the push provided by the battery.
current	The flow of charge around a circuit.
resistance	Measure of how easy it is for the current to flow around a circuit.
attract	The invisible magnetic force between the magnets will pull the poles together.
repel	The invisible magnetic force between the magnets will force the poles away from each other.
independant variable	This is the variable that we change.
dependant variable	This is the variable that we measure.
control	This is the variable that we keep the same.

In a parallel circuit, the components are connected side by side. This gives the current several different paths for it to flow around. If one bulb blows, the other bulbs will remain lit as the circuit is still complete.

In a series circuit, the components are connected end to end in a loop.
If one bulb breaks, the whole circuit will go out and none of the bulbs will light as the circuit is no longer complete.

Ammeters are always connected in series to measure the current of a circuit. Voltmeters are always connected in parallel to measure the voltage of a circuit.



Electrical Circuits

Electrical circuits are often represented by circuit diagrams. Rather than spending lots of time creating very artistic drawings of circuits, we use standard symbols that are much simpler, easier to draw and easier to interpret.
The symbols represent components such as a bulb, ammeter and voltmeter that can be used in a circuit diagram.

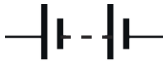

bulb



ammeter



voltmeter

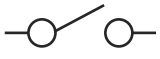

battery


cell


buzzer



motor


open switch


closed switch

Resistance

Resistance is a measure of **how easy** it is for the current to flow around a circuit.

The **higher the resistance**, the **lower the current** that will flow around the circuit. The **higher the current** flow around the circuit, the **lower the resistance**.

Resistance is measured in ohms. (Ω)

Modelling Circuits

Scientists often use models to help them to explain difficult concepts.
They help scientists to think about an idea and explain what is happening, to other scientists. Some models, however, are better than others.

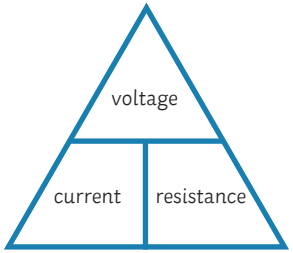
Batteries

Batteries store **chemical energy** and transform it into electrical energy. This is how batteries are able to power our electronic devices like mobile phones, torches and a lot more.
Batteries contain an **electrolyte** and **two electrodes**. One of the electrodes is **positively charged** and the other **negatively charged**. A chemical reaction between the two electrodes creates a flow of electrical energy to the device.



Calculating Resistance

resistance = voltage ÷ current



Voltage

The voltage of a battery tells us how much **energy** it provides to the components in the circuit.
It also tells us how **hard** a battery pushes the negative charges in a circuit: the bigger the voltage, the **bigger** the push.
The strength of the push provided by the battery is called the **voltage**.

Voltage is measured in volts (V).



Current

Current is the flow of **charge** around a circuit. The faster the **charge** flow, the higher the current.
Current is measured in **amps** using an **ammeter** (always in series).

The current is the **same** everywhere in a series circuit. It doesn't matter where you put the ammeter, it will always give you the same reading. The more cells or batteries you add, the greater the current. Current is not used up.

Conductors have low resistance as they allow a flow of charges to pass through them.

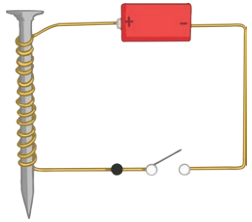
Insulators have high resistance as it is difficult for a flow of charges to pass through them.



Making an Electromagnet: Core Practical

Equipment

1 power pack, 1 large nail, 30cm of insulated copper wire, 2 crocodile clips, 2 wires and 30 paperclips.



Method

Step 1 – Collect the equipment.
Step 2 – Place crocodile clips on two of the wires.
Step 3 – Attach the opposite end of each wire onto the power pack.
Step 4 – Wrap the copper wire around the nail until you reach the required amount of coils. Use a pair of wire strippers to remove some of the insultation. Leave 2cm of exposed wire at each end of the copper wire.
Step 5 – Attach the crocodile clips to the exposed wire.
Step 6 – Lay the paperclips on the bench and hold the insulated wire either side of the nail.
Step 7 – Hover the nail over the paperclips and record in your table how many paperclips are attracted to the nail.

Magnetism

Magnetism is an invisible force.

All magnets have a **north** and a **south pole**.

Like poles **repel**. This means that the two poles will push away from each other.



Opposite poles **attract**. This means that the invisible magnetic force between the magnets will pull the poles together.



Not all metals are magnetic. **Iron**, **nickel** and **cobalt** are the three magnetic metals.

The earth has a **magnetic inner** core made of iron.

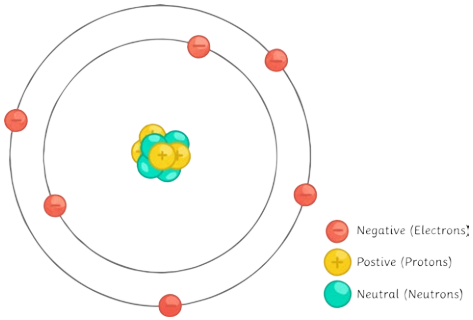


There are two kinds of charge: positive (+) and negative (-).

All objects are made up of atoms. Each atom contains three different types of particle.

A **positive particle (proton)**, a **negative particle (electron)** and a **particle with no charge** (**neutron**).

Atoms contain the same number of protons and electrons. The positive and negative charges cancel out and so an atom has no overall charge.



Static electricity occurs when an object either **loses** or **gains** a **negative charge**.

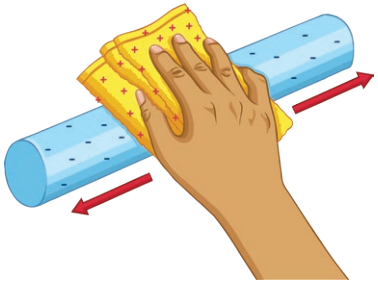
We say that objects that lose negative charges become **positively (+) charged** overall.

We say that objects that gain negative charges have a **negative (-) charge** overall.

When a polythene strip is rubbed with a cloth, the **negative charges** move from the cloth to the strip.

The **strip becomes negative** because it has more negative charges than it needs to cancel out the **positive charges**.

The **cloth becomes positive** because it has less negative charges than it needs.



Electromagnet

When electrical charges flow in a wire, a magnetic field is formed around the wire.
The larger the current, the stronger the magnetic field.
When a current stops flowing, there is no magnetic field.
The more coils of wire around the nail, the stronger the magnetic field and the more paperclips it can hold.

Static Electricity

Static Electricity: Dangerous or Just a Nuisance?

Petrol tankers travelling on the roads.
Using a mobile phone on a petrol station forecourt.
Cleaning an oil tanker.
Dust attracted to insulators e.g. the TV screen.
Clothes made from synthetic cloth cling to each other.

Van de Graaff Generator

Electrons are transferred onto the dome so the dome gets an electrical charge.
When we touch the dome, the **electrical** charge is passed to us.
We stand in a plastic tray and this is because plastic is an **insulator**.
The electrical charge cannot flow through our bodies and feet to the earth.
Our hair stands up because each strand is negatively charged and the same charges **repel**.
When the electrical charge jumps between the two domes it makes a **spark**.
This can light a **flammable** gas such as methane.

